Modelling salmon spawning surveys as arrivals and residence, an advance from area under the curve estimates.

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Abstract

The abundance of spawning salmon is a fundamental variable for salmon management, but current methods provide estimates without uncertainty and biased from assumptions of residence time (to correct for observing a fish in repeated surveys). We used a new model – coded as a multi-level, non-linear, Bayesian regression – to reveal annual abundance, the timing and spread of entry to the spawning grounds, and spawner residence time. The model was challenged with simulated data to demonstrate robustness, and challenged with data from 291 surveys (25 years) of wild Sockeye Salmon (*Oncorhynchus nerka*) spawning in the Okanagan River upstream of Osoyoos Lake. Abundance estimates appear to be more accurate than conventional. While entry timing and spread varied considerably between years, residence was relatively constant. We also discovered that exit spread varies by year. We foresee this model including more ecology, such as effects from spawner biometrics and habitat variables.

Keywords: salmon, Oncorhynchus, spawner, spawner survey, Osoyoos Lake, Okanagan River, Columbia River

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1. Introduction

1.1. The importance of spawner surveys

Spawner abundance is a key variable in salmon ecology. Functions relating spawners to their offspring that return as adults ("stock" and "recruits") are

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fundamental for management of salmon fisheries and salmon habitats (Ricker, 1975; Hilborn and Walters, 1992).

Estimates of the number of Sockeye Salmon (*Oncorhynchus nerka* Walbaum 1792) spawners in a river or a lake beach involve repeated surveys during the spawning period. Methods vary; conventionally: visual counts while boating, snorkeling, walking, or flying in a helicopter or small airplane; and recently, analyzing video from drones. There are many other methods to estimate spawner abundance: mark-recapture, conductivity sensors, conductivity sensors, counting fences, eDNA, and combinations of methods. Complications include distinguishing hatchery fish from wild, and separating anadromous Sockeye from non-anadromous Kokanee (sub-species of *O. nerka*).

1.2. AUC: area under the curve

The conventional analysis of a survey of sockeye spawners determines **AUC**, the integral of spawner count over a number of surveys (units: fish×days). Spawners have a **residence** time on spawning grounds (aka, survey life), so a spawner will be observed more than once in a series of surveys. The estimate for abundance of spawners divides AUC by residence to correct for repeated observations. That estimate of abundance is then proportional to 1/residence. Using an assumed value for residence time changes the absolute number of spawners in each survey to an index of relative abundance Askey et al. (2023). There are two approaches to estimating AUC:

1. **non-parametric**: where "curve" means a trapezoid of linear interpolations of counts between survey dates [English et al. (1992); Irvine et al. (1992);]. Prior knowledge about first and last presence of spawners (assumed dates for bounding zeros) is required. This is a robust estimator (ref?) and efforts have been made to estimate the precision of AUC from precision of observations (Askey et al., 2023; Korman et al., 2002; Millar et al., 2012; Parken et al., 2003; Parsons and Skalski, 2010). Essentially, AUC determines the maximum of a cumulative probability distribution of spawner number estimated by linear segments.
2. **parametric**: a probability density distribution, typically Gaussian, is fitted to the time series of survey estimates in a year Hilborn et al. (1999). This has been extended by multi-level regression to analyze multiple years simultaneously (Adkison and Su, 2001; Labelle et al., 2021; Su et al., 2001). Then AUC is the area under that curve. Additional parameters and different distributions can provide truncated, skewed, or flattened variants.

Both require an assumed value for residence to estimate spawner abundance. Estimates for residence are determined independently of spawner surveys, using methods such as a spawner fence or recapture of fish (after death) tagged upon entry (Askey et al., 2023; Berghe and Gross, 1986; Cormack, 1964; Fukushima and Smoker, 1997; Korman et al., 2002; Lady and Skalski, 1998; Liao, 1994;

Manske and Schwarz, 2000; Neilson and Geen, 1981; Perrin and Irvine, 1990; Semenchenko, 1993; Zaporozhets and Zaporozhets, 2021). also Bue et al. (1998) and English et al. (1992) and Thomason and Jones (1984).

1.3. Objectives

An advance from AUC by a new model will estimate **residence** directly, as well as **timing** (mean day) and **spread** (standard deviation) of spawner entry. This will improve accuracy of **run** estimates (spawner abundance) and yield precisions for those estimates. Analyzing multiple years of surveys simultaneously will provide annual estimates of parameters, improved by using groups in a multilevel regression.

2. Background

Our analysis was confined to the population of wild Sockeye Salmon that spawn in October and November in the Okanagan River (Figure 1) upstream of Osoyoos Lake. After emerging from spawning gravel, mid-April to mid-May, they spend one or two summers (rarely three) as parr in the North Basin of Osoyoos Lake before an ontological shift (near autumnal equinox) to pre-smolts and subsequent migration the following spring as smolts through the reservoirs (with abundant predators), spillways, and smolt passage tunnels of ten hydroelectric dams to reach the Pacific Ocean via the Columbia River.

After one to three years (rarely more) of marine growth, adults return to the Columbia River in late summer and attempt the transit of ten dams in a 950 km freshwater migration, before entering Osoyoos Lake and subsequently spawning yet further upstream. Ascending the Okanagan River from the Columbia River is increasingly compromised by high river temperatures (Isaak et al., 2011; Hyatt et al., 2020; Bailey and Freshwater, 2023; Martins et al., 2012; Major and Mighell, 1967; Quinn et al., 1996).

Spawner estimates for Osoyoos sockeye have historically been based on trapezoidal AUC, with an assumed residence time of 11 days (**?**). However, residence time was never assessed for this population directly. Therefore, the selected residence time was based on the estimate for the early Stuart sockeye stock (Perrin and Irvine, 1990) because that stock has similar average body size, migration distance, and migratory timing (**?**).

3. Data

3.0.1. Enumeration surveys

Spawner enumeration surveys have been conducted in the main sockeye spawning area—also known as the Index section—of the Okanagan river between Osoyoos and Vasseau lakes (Figure 1) since 2000. The Index section consists of the

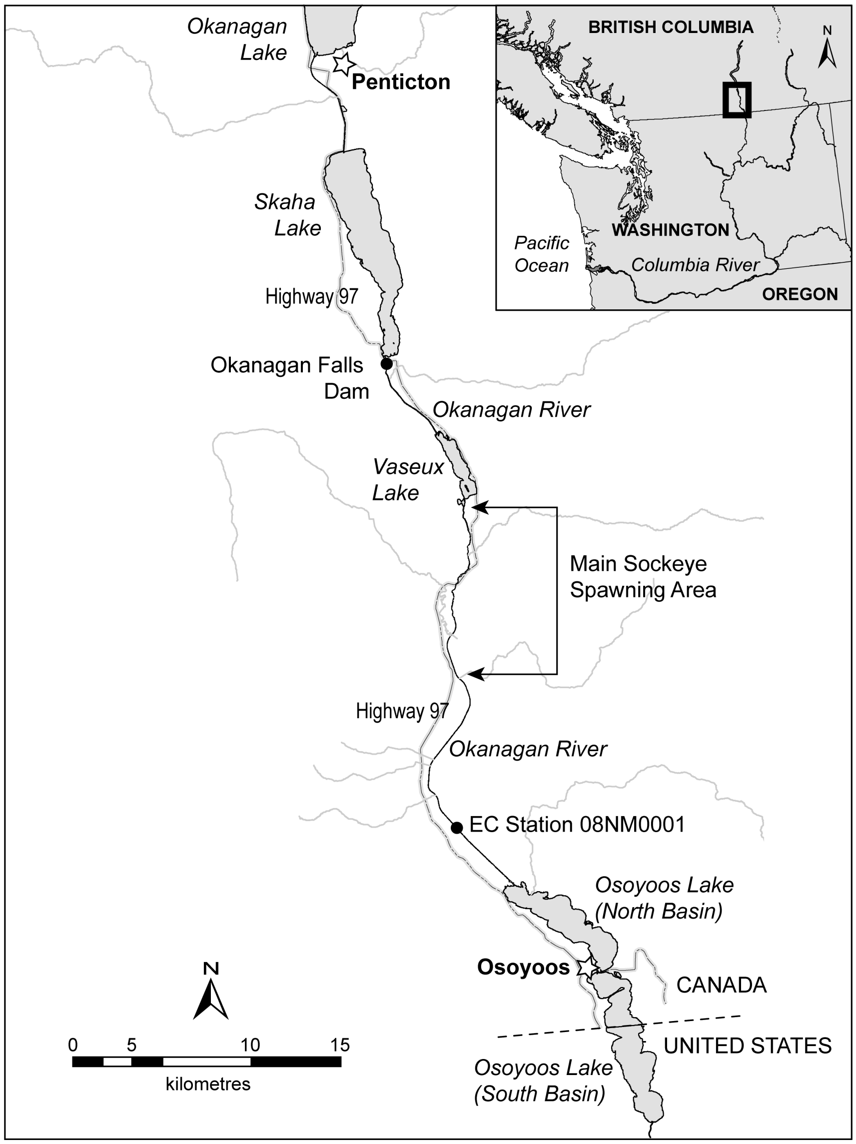

Figure 1: The spawning grounds for Sockeye Salmon population that rear as parr in the North Basin of Osoyoos Lake are ~20 to 30 km upstream, in the Okanagan River below McIntyre Dam at the outflow of Vaseux Lake. River flows that affect spawners, eggs, and parr are monitored by Environment Canada (EC) at the site indicated. Map by Braden Judson, Pacific Biological Station.

section from Deer park to vertical drop structure 12 (VDS12), which is at the Oliver bridge. All together, surveys have been conducted on 291 days, ranging between 9 and 19 surveys per year. All surveys occurred between August 28 and November 24.

Spawner enumeration survey methods are described in detail in (**?**). They are conducted on a weekly basis during the initial migration phase, then increased to a three-day interval during the peak spawning period (generally October-November). A three to four person crew performs visual surveys of the Index section from a boat. Two observers focus on counting live sockeye (holding and spawning), one on each side of the middle of the boat, while another one or two observers count dead sockeye and other species (Chinook, kokanee, and rainbow trout) from the front of the boat. In addition, side channels throughout the Index section are walked and fish are enumerated. Since 2012, enumeration of each individual reach within the Index section is available, however, we have used the summed counts across all reaches, because this is the resolution of data that is available from 2000–2011. Limited spawning for the Osoyoos population also occurs in the Okanagan river downstream of the Index section, and in the section between Vaseaux Lake and Okanagan Falls Dam. However, we have excluded these sections from our analysis as including them would require additional complexity in our model to account for potential variation in timing. It would also require that we account for the relative abundances across river sections, that vary through time. This sort of extension is a logical next step, but is not necessary for demonstrating our method, which is the goal of this manuscript.

Although the survey crews count dead sockeye, the dead surveys do not provide a consistent estimate of the accumulation of dead spawners that could be used in our model. Dead fish are enumerated but not marked or removed and so can potentially be counted multiple times across survey dates. However, there is a separate deadpitch crew that collects biosamples multiple times over the course of the spawning period, but at a lower frequency than the live counts are conducted. When doing so, this crew removes the sampled fish. Therefore, the counts of dead fish by the enumeration crew represent accumulated dead spawners, with infrequent removal, that is not easily accounted for in a model.

4. Methods

4.1. Problem: residence time

Spawners reside on spawning ground between repeated surveys, so a spawner may be counted repeatedly. The conventional estimate of abundance uses a value of residence to correct AUC for repeated observations of spawners. Residence is an assumed value, however, derived independently of spawner surveys. Can we do better?

4.2. Solution: spawner entry and exit

The timing and spread of spawners' entry to a spawning ground is reasonably modeled as a Gaussian (normal) probability density distribution (PDD). If residence is constant (ignoring sex, age, size, condition, injury, history, and genetics), then the PDD of spawner exits is the same as the PDD of entry, just delayed. The number of live spawners in a survey day is then the cumulative distribution of entries to that day minus the cumulative distribution of exits. This model is sufficient to estimate runs and residence simultaneously, but assumes all spawners are identical. In what follows, the assumption of fixed residence is relaxed to allow exit spread differ from entry spread. for a distribution of residence times, but the model is still based on the assumption that all spawners are identical.

If available, counts of newly dead spawners provide further information to estimate residence. Newly dead spawners accumulate between surveys, so the counts depend on survey intervals: the number of newly dead spawners is exits (deaths) cumulative to this survey day, minus deaths cumulative to the preceding survey. This assumes that dead spawners are marked by tags, by cutting off tails, or "dead pitch" (removal from the river bank) to ensure they are only counted once. Some dead spawners will drift away, not counted, necessitating a factor to estimate fraction of dead observed.

4.3. Model

We modelled spawner abundance based on observed spawner counts using a series of hierarchical Bayesian models fit in the Stan programming language (**?**). These models varied in complexity and the degree to which they allowed for interannual variation in the arrival and exit processes. However, all models follow the general set of processes shown in Figure 2. Live spawners counts are integers and can include zero. Therefore, we used a Negative Binomial distribution for the observation model:

$$N_{y,d}^{\text{live}} \sim \text{NegBinomial}(\mu_{y,d}^{\text{live}}, \phi^{\text{live}}) \tag{1}$$

where $\mu_{y,d}^{\text{live}}$ is the expected number of live spawners present in year y on day d, and ϕ^{live} captures variability around $\mu_{y,d}^{\text{live}}$, including both observation error and natural variation that is not explicitly modelled. We assume that this expected number of spawners $\mu_{y,d}^{\text{live}}$ is a function of three unobserved (i.e., latent) quantities that are estimated in our model: the proportion of spawners that have arrived by day d, $p_{y,d}^{\text{arrived}}$; the proportion of those that have exited the live population, $p_{y,d}^{\text{exited}}$ (i.e., died or left the spawning grounds), and total spawner abundance in that year, N_y^{spawners}:

$$\mu_{y,d}^{\text{live}} \approx p_{y,d}^{\text{arrived}} \times (1 - p_{y,d}^{\text{exited}}) \times N_y^{\text{spawners}}. \tag{2}$$

We assume that the spawners arrive over a range of days that follows a normal distribution, reside there for some days of spawning activity, and then die or leave the spawning grounds, thereby exiting the live counts following a second normal distribution that lags the arrival distribution. Thus, the proportion of spawners that have arrived by a given date d in year y follows a cumulative normal distribution:

$$p_{y,d}^{\text{arrived}} = \Phi\left(\frac{d - \mu_y^{\text{arrival}}}{\sigma_y^{\text{arrival}}}\right). \tag{3}$$

Likewise, the proportion of arrived spawners that have exited the live count by a given date d in year y follows a second cumulative normal distribution:

$$p_{y,d}^{\text{exited}} = \Phi\left(\frac{d - (\mu_y^{\text{arrival}} + \mu_y^{\text{exit lag}})}{\sigma_y^{\text{exit}}}\right), \tag{4}$$

where $\mu_y^{\text{exit lag}}$ defines the number of days by which the mean exit date lags behind the mean arrival date μ_y^{arrival}. Total spawner abundance, N_y^{spawners}, in each year is a primary variable of interest and is estimated by the model as the scaling factor that links the estimated arrival and exit processes to the expected live spawners, $mu_{y,d}^{\text{live}}$. We set a prior on spawner abundance as $log(N_y^{\text{spawners}}) \sim \text{Normal}(10, 1)$, which is necessary because it is an estimated parameter. This weakly informative prior is consistent with the range of log-transformed spawner abundances previously estimated for this population using trapezoidal AUC, but does not constrain estimates to match those values.

To prevent numerical instability when equation Equation 2 approaches zero, we use a log-transformed approximation of the same form:

$$\mu_{y,d}^{\text{live}} = \exp\left(\log\left(1 + 10^4 \times p_{y,d}^{\text{arrived}} \times (1 - p_{y,d}^{\text{exited}})\right) - \log(10^4)\right) \times N_y^{\text{spawners}} \tag{5}$$

where the factor 10^4 serves as a scaling factor that prevents changes in near-zero values from having an abrupt influence on $\mu_{y,d}^{\text{live}}$.

In the equations above, we specified μ_y^{arrival}, $\sigma_y^{\text{arrival}}$, $\mu_y^{\text{exit lag}}$ and σ_y^{exit} using subscript y to indicate that they vary from year-to-year. However, it is not clear that all of these processes should vary annually, so we compared a series of models with different combinations of varying and constant versions of these parameters (summarised in Figure 3 A) and evaluated their fit to withheld data, as described in a subsequent section.

The time-varying versions of these parameters were estimated using a hierarchical structure, where yearly values were drawn from a hyperdistribution (e.g., $\mu_y^{\text{arrival}} \sim \text{Normal}(\mu_\mu^{\text{arrival}}, \mu_\sigma^{\text{arrival}})$) defined by the multiyear mean and year-to-year standard deviation. We set a prior for the arrival day group mean as $\mu_\mu^{\text{arrival}} \sim \text{Normal}(40, 5)$, corresponding to a mean of approximately 45 % of the typical spawning survey period, with a prior interval spanning 34–57. This reflects our expectation that peak arrivals occur during the early to mid portion of the survey window. We used the prior $\mu_{\sigma}^{\text{arrival}}$ Exponential(2) to represent a moderate level of expected year-to-year variation in μ_{y}^{arrival}. We modelled $\sigma_{y}^{\text{arrival}}$, $\mu_{y}^{\text{exit lag}}$ and σ_{y}^{exit} using priors of the form $\text{Exponential}(\lambda) + 1$ to ensure that parameters were strictly positive and bounded away from zero following the biological constraint that spawners vary in their arrival and exit timing, and that they typically remain on the spawning grounds for at least one day. We set priors for $\sigma_{\mu}^{\text{arrival}}$ and $\sigma_{\mu}^{\text{exit}}$ as $\sim \text{Normal}(\log(7), 0.2)$, corresponding to an expectation that arrival and exit timing should have similar spread. This corresponds to an expected range of durations between 6 and 11 days. We set the prior for $\mu_{\mu}^{\text{exit lag}}$ as $\sim \text{Normal}(\log(10), 0.3)$, aligning with the previously used residence time of approximately 11 days, and with likely values ranging from 7 to 19 days. We used the same Exponential(10) prior for $\mu_{\sigma}^{\text{spread}}, \mu_{\sigma}^{\text{exit}}$, and $\mu_{\sigma}^{\text{exit lag}}$ to represent a moderate level of expected year-to-year variation in these parameters on the log-transformed scale.

In the models where we use constant versions of the timing parameters, we drop the subscript y and they are denoted as: μ^{arrival}, σ^{arrival}, $\mu^{\text{exit lag}}$, and σ^{exit}. In these cases, we use the prior for the group level μ and omit the group-level variance parameter, thus removing the hierarchical structure. The full model equations including priors are presented in Equation 8.

4.3.1. Diagnostics

An essential concept in Bayesian statistics is that the data is known but the best model for that data is unknown McElreath (2016) and always a vast simplification of reality Jaynes (2003). Metrics to compare and rank competing models are required. In ecological systems, a superabundance of habitat indicators as covariates means over-fitting is a problem. Leave-one-out (LOO) statistics[2], where each point is predicted while omitted from fitting (cross-validations), provides diagnostics for over-fitting. We used ELPD, the sum of expected log predictive probability density for each of n omitted observation's prediction, and $\text{ELPD}_{SE} = \text{sd}(\text{ELPD})/\sqrt{(}n)$ the standard error of those probabilities Vehtari et al. (2017).

4.4. Mean residence

Although $\mu_{y}^{\text{exit lag}}$ defines the average delay between arrival and exit distributions, it does not equal the mean residence time. This is because residence time depends on the interaction of both arrival and exit processes and is not directly estimated by any single parameter of the model. Importantly, whenever $\sigma_{y}^{\text{arrival}} \neq \sigma_{y}^{\text{exit}}$, residence time cannot be the same for all individuals, and instead varies depending on when the fish arrive. We can, however, calculate a

[2]see R package LOO https://cran.r-project.org/web/packages/loo/loo.pdf and LOO Package Glossary https://mc-stan.org/loo/reference/loo-glossary.html.

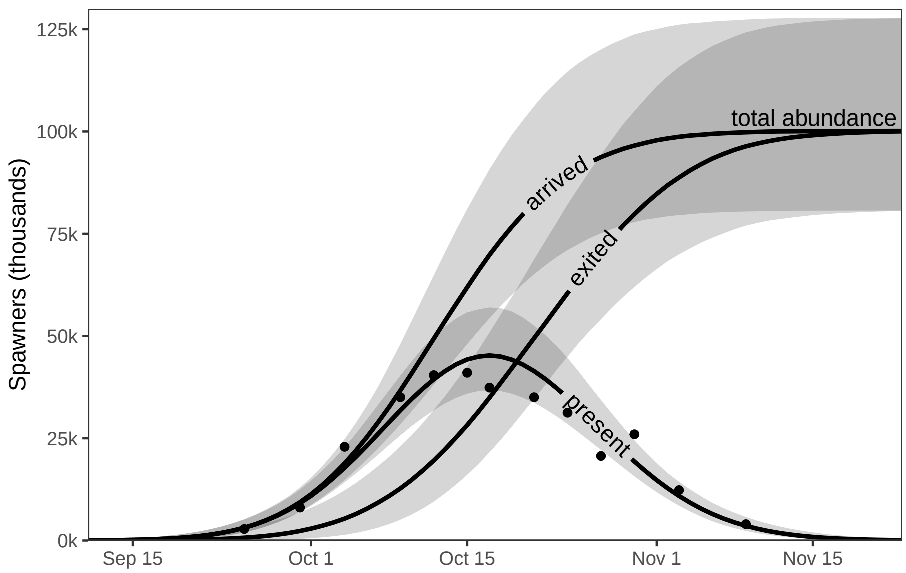

Figure 2: Example of how the model estimates total spawner abundance, arrival timing, arrival spread, mean residence, and exit spread based on the observed counts (points) of spawners on multiple days. The number of spawners present on a given day is the difference between accumulated arrivals and accumulated exits. We assume that spawner counts are imperfect observations of the true number, and so indiviual points vary from the estimated presence value. The total abundance of spawners is estimated as the total number that have arrived. All model parameters are estimated with uncertainty — lines show the posterior median and bands show the 95% credible intervals. In the case of arrived and exited numbers, uncertainty increases with abundance because uncertainty from the timing and spread of arrival or exit is multiplied by uncertainty from abundance.

mean residence time in each year based on the sum of estimated number of live spawners across all days in the spawning period, D, divided by the estimated total spawner abundance:

$$\mu_y^{\text{residence}} = \frac{\sum_{d=1}^{D} \left(p_{y,d}^{\text{arrived}} \times (1 - p_{y,d}^{\text{exited}}) \times N_y^{\text{spawners}} \right)}{N_y^{\text{spawners}}}. \tag{6}$$

This is the equivalent of back calculating mean residence time from the area under the curve, which is possible because we have directly estimated N_y^{spawners}. However, because N_y^{spawners} appears in both the numerator and the denominator of Equation 6, we can simplify this equation as:

$$\mu_y^{\text{residence}} = \sum_{d=1}^{D} \left(p_{y,d}^{\text{arrived}} \times (1 - p_{y,d}^{\text{exited}}) \right). \tag{7}$$

4.5. Simulations

We challenged this model with simulated observations to ensure robust and reliable results. Parameters were sampled randomly from prior PDDs. Simulated surveys were deliberately poor: the number of surveys each year was random uniform between 6 and 10, and survey dates within years random uniform but separated by at least 1 day. The resulting "true" data, but with multiplicative normal sampling error (% accuracy), was the "observed" data subsequently fitted by the multi-level regression coded in Stan. The scenarios explored fits (see Stan Model) with and without surveys of newly dead spawners. Despite noisy observations and gaps between surveys, parameters were recovered accurately from Stan model, with the caveat that the model used to fit the simulated data was the model that created it. Specifically, this procedure correctly estimated residence over a wide range. The simulations also generated indices of relative abundance of spawners, such as AUC as Gaussian and as linear interpolations (a trapezoid), peak live, and peak live plus dead (PLD). These were compared to the fitted run size to indicate reliability.

5. Results

> outline: Simulations recapture residence. why critical. Tables * descriptions and fits (ELPD, AIC,) comparison of Stan models
> * parameter estimates and statistics for best model (so far).
> Plots
> * example of model with CLs to entry and exit CPDs, observed datapoints, and CLs for predicted curve. Just one year. * timing, spread, residence by years, with CLs. Fancier the better.

5.1. Model selection and convergence

Based on ELPD statistics (Figure 3 B) we selected model 7 as the best from those explored. This model included year-to-year variation in spawner abundance, arrival timing, arrival spread, and exit spread, but assumed that residence time and observation error is constant across years. Model 8, which also allowed residence time to vary across years had an equivalent ELPD (ΔELPD = -1.5, SE = 1.7).

The effective number of parameters (p_{loo}) increased when arrival timing variation was introduced (model 1 vs. 2), and also when any of arrival spread, exit spread, or residence were allowed to vary (models 3 to 8; Figure 3 C). However, all models with variation in any or all of these parameters (models 3 to 8) had similar numbers of effective parameters.

Convergence was good for model 7; all parameters had $\hat{R}$ values ≤ 1.003 and effective sample sizes ≥ 695. Allowing residence time to vary across years (model 8), resulted in some minor convergence issues; 0.9% of parameters had $\hat{R} > 1.01$, and the highest $\hat{R}$ was 1.01. Model 8 also had 1.3% of its parameters with effective sample sizes < 400, the lowest being 229. The reason that model 8 exhibited minor convergence issues is likely because of a posterior correlation between residence time and the other timing parameters, which are only partially identifiable from the observed spawner counts. Thus, we have selected the simpler model (model 7), which has equivalent predictive power and which does not have convergence issues.

5.2. Model fit to observed counts

Our expected counts correspond well with the observed counts Figure 4 and Figure S1. Variation between the observed and expected is quantified in our estimate of spawner count overdispersion (ϕ^{live}), which had a posterior median of 6.9 (95% CI: 5.5–8.5).

5.3. Spawning timing

Arrival timing (μ_y^{arrival}) varied across years (Figure 5 A), with posterior medians ranging from October 10 to October 15. Arrival spread ($\sigma_y^{\text{arrival}}$) varied across years (Figure 5 B), with posterior medians ranging from 6 to 13 days. Exit spread (σ_y^{exit}) tended to be lower than arrival spread (Figure 5 C), with posterior medians ranging from 5 to 11 days. Exit lag ($\mu^{\text{exit lag}}$) was estimated as 8.8 days (95% CI; 6.4–11.6).

Together, the timing parameters result in year-to-year variation in the mean residence time (Figure 5 D). The posterior median across years is 10.5 days, but it ranges from 9.8 to 11.5 days.-

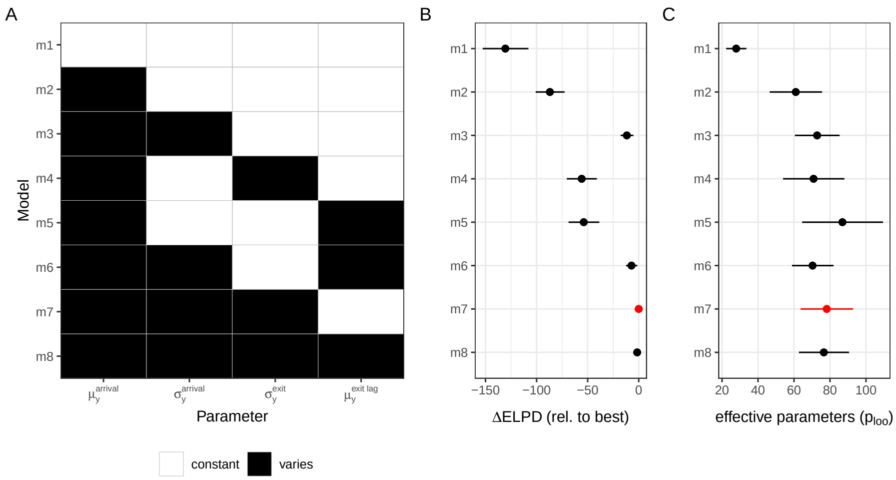

Figure 3: Model summary and comparison. *Panel A* shows which parameters were held constant (white) versus varied (black) across years in each of the candidate models (rows). *Panel B* shows the relative expected log predictive density (ΔELPD) compared to the model with the highest out-of-sample predictive accuracy. *Panel C* shows the effective number of parameters (p_{loo}). The model with the highest out-of-sample predictive accuracy is highlighted in red in panels B and C. Points and error bars (B and C) represent means and standard errors.

5.4. Total spawner abundance

Estimated posterior median annual spawner abundances (N_y^{spawners}) varied from as low as 3700 in 2002 to as high as 176,000 in 2010. Since 2008, spawner abundance has alternated between low and high levels in a two-year cycle.

Our estimated annual abundances are, on average, slightly higher than those from the traditional trapezoidal AUC estimate, with an assumed 11 day residence time. The median difference across all years is 4 % (inter-annual range: -10.6–12), which is a smaller difference compared to the 19.8% increase that we would expect if we were to apply our estimated residence time (8.8 days) to the traditional method. In addition, while our spawner abundance estimate tends to be higher, this is only the case in 16% of the 25 years and the traditional estimate falls within the 66% credible interval of our model prediction in all years.

6. Discussion

outline: General- new model is successful.
Run size estimates are more accurate than AUC.
Which parameters (models) were important (annual spread and timing), which not (varying residence, autocorrelation of surveys)? Res-

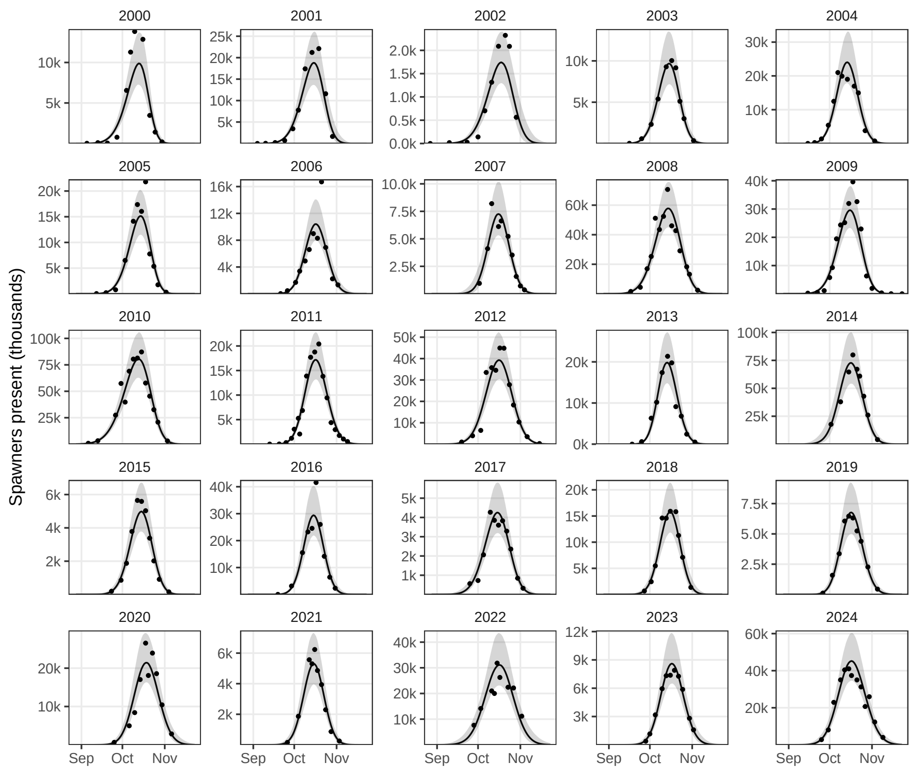

Figure 4: Observed versus estimated spawners present over the spawning period in each year (panels). Points show the observed counts, the lines and shaded bands show the posterior median the 95% credible intervals. Y-axis limits vary across panels to emphasize model fit rather than variation across years.

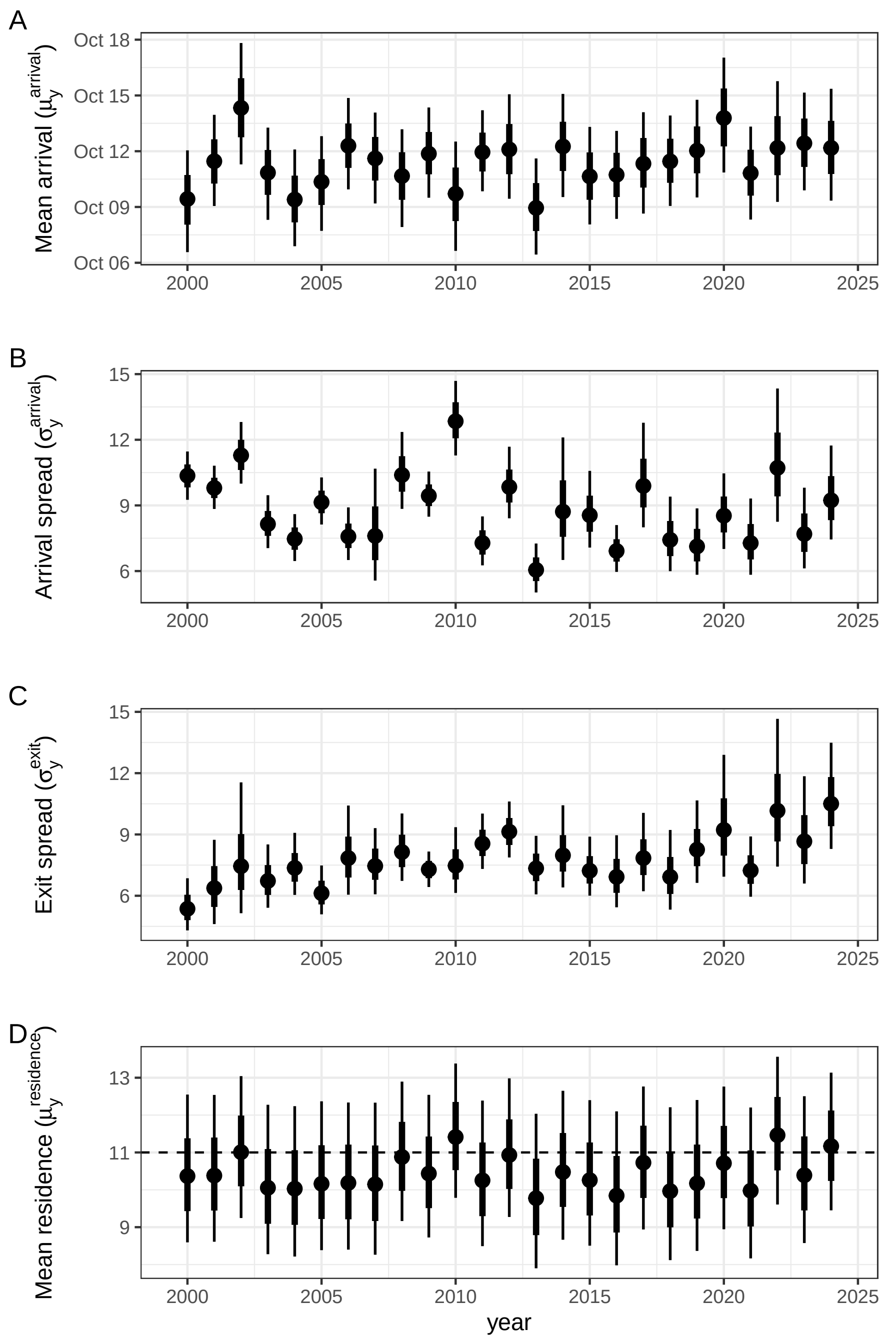

Figure 5: Posterior distributions of mean arrival (μ_y^{arrival}; A), arrival spread ($\sigma_y^{\mathrm{arrival}}$; B), exit spread ($\sigma_y^{\mathrm{exit}}$; C), and the resulting mean residence time ($\mu_y^{\mathrm{residence}}$; D). The dashed line in panel D shows the previously assumed 11 day residence time. Points and error bars show the posterior median and the 66% and 95% credible intervals.

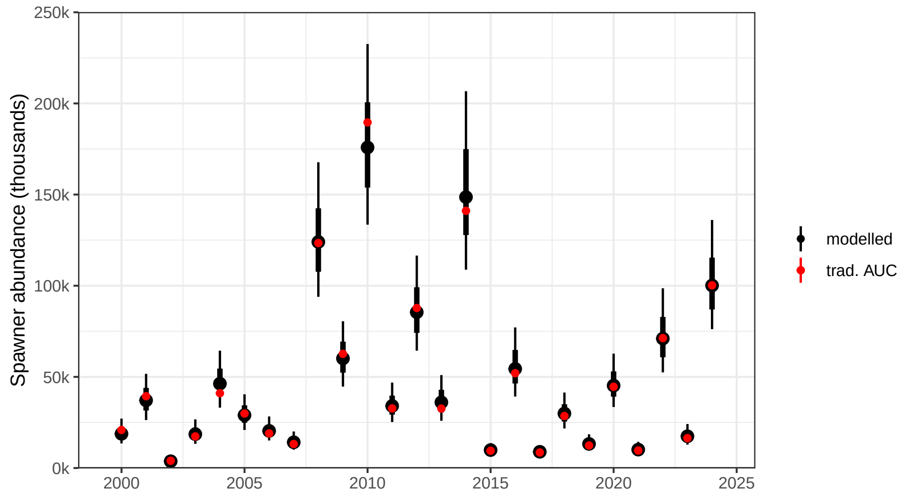

Figure 6: Posterior distributions of total spawning abundance by year (black) compared to the point estimate based on traditional trapezoidal AUC with an assumed 11 day residence (red). Points and error bars show the posterior median and the 66% and 95% credible intervals.

> idence varies little, so conventional is a reliable relative index (?) (but I need to understand how you changed shape of exit).
>
> How to accommodate biometrics: if early entries (big females?) stay longer,. Scope for covariates: biometrics and habitat, parameter correlations,. Events like Testalinden. scope for more detail: model to predict counts by segment. Value of (good) surveys of newly dead. (?? Value of analyzing all data simultaneously. possible extensions: * autocorrelation in spawner runs (ages, SAR,); * models for entire life history (add parr, smolts, returns (SAR), prespawning mort)

6.1. Residence and exit spread.

The attempt to estimate residence directly from spawner surveys was successful. For this species and place, the estimate was ?? days ($ = ??). This is approximately twice the spread of entry, implying nearly half of the spawners enter before newly-dead are abundant. Our estimate suggests that residence time is likely lower than the value (11 days) previously used to correct for AUC (**?**), which sits at the 94^th percentile of our distribution. Furthermore, we identified annual variability in exit spread that is related to changes in residence. Our interpretation is that residence time is a function of date of entry, likely because early entrants stay longer than late entrants, a pattern identified by REFERENCES and ascribed to age and sex. Residence as a linear function of entry changes exit spread compared to entry spread:

$$\mathcal{N}(x, \mu, \sigma) + (a + bx) = \mathcal{N}(x, \mu + a, b\sigma)$$

where a is the mean residence time (constant), and b where $(0 < b < 1)$ describes how much the exit spread shrinks from entry spread. If entry spread $= 5$ days and $b = 0.8$ then exit spread $= 4$. We determined exit spread directly rather than determining b because the the underlying ecology will be more complicated than a single, linear, parameter. Inspection of Figure 5 reveals exit spread is not correlated with entry spread, but does share a common temporal trend with entry timing (recently later and longer). That exit spread varies is an intriguing discovery awaiting explanation.

6.2. Annual run size

Our model estimates precision for annual run size (spawner abundance). This is valuable as regression weights for subsequent application (parr from spawners, returns from spawners). Estimated as a negative binomial coefficient, it is a multiplicative factor (% accuracy) of ??($\sigma =$??). As a coefficient of variation (σ/μ), this varied from ??($\sigma =$??) in 2000 to??($\sigma =$??) in 2000 (Figure 6). Note this precision includes model error as well as observation error.

Because our estimate of mean residence time is similar to value used to correct convetional AUC estimates, the 95% uncertainty intervals for run estimates from this analysis overlap most of the conventional estimtes. The exception is determining that conventional estimates (red dots in Figure 6) for years 2016 to 2018 might be substantial underestimates. A severe overestimate in 2011 was noted, but investigation revealed the conventional estimate had been increased, arbitrarily, to better match spawner passage estimates from Columbia River dams. Conventional estimates could not be accessed for years after 2019. Although the results for Osoyoos Lake Sockeye, and the results from simulations, provide initial confidence in the accuracy of these run estimates, complete confidence awaits resolution of the 2016-2018 discrepancies and results from different salmon populations.

6.3. Entry timing and spread

Timing and spread of spawner entry are ecologically informative parameters not previously extracted from spawner surveys. Timing overall was day 285($\sigma =$??) and ranged from day 282 in 2013 to 288 in 2002, less than a week. Although recent timings (2016–2024) do not have any of the earliest timings (2000–2015) this is not a linear trend ($r^2 = 0.00$).

Entry spread was 285($\sigma =$??) days, ranged from 6 days (2013) to 11 days (2002). Although the estimate of 12.7 days for 2010 is the widest estimate, this is associated with the 2010 June 13 Testalinden Creek dam failure that disgorged silt and toxins to the Okanagan River downstream of the spawning region, but affecting Osoyoos Lake and the Okanagan River below it. Entry spread is the only parameter apparently affected by that event; how that affected spawners, if it did, has yet to be determined.

6.4. Relative importance of parameters

Which parameters (models) were important: annual spread and timing; and which not: varying residence, autocorrelation of surveys,?

6.5. Adding covariates: biometrics and habitat indices

This model is the basis for expanding the information applied to analyses of spawner surveys. Perrin and Irvine Perrin and Irvine (1990) identified a suite of potential covariates for residence, including spawner density, sex, body size, body size by sex (females stay longer, to guard redds, than males), sex ratio, run timing (a year with late timing may have shorter residence), time of arrival, flow, and water temperature; concluding *Since residence can vary among years within a stream, and among streams within a year, residence estimates should not be transferred among streams or years. This can produce serious errors in the escapement estimate. Residence should be determined on a site specific basis each time the AUC estimate is used to estimate escapement* [replacing 'survey life' with 'residence'].

Spawner biometrics affect entry timing, entry spread, and residence (thus exit spread): *In general, older and larger wild Atlantic salmon return to freshwater to spawn earlier in the season than younger and smaller salmon, and female salmon return earlier in the season than males* Miettinen et al. (2024).

EXACTLY HOW SHOULD BIOMETRIC AND HABITAT DATA BE INCLUDED, AND VALUE THEREOF BE DETERMINED?

Spawner surveys collect length, sex, and age (otoliths and/or scales) from dead salmon, with varying precisions for mean length at age by sex. Typically this can be augmented with biometrics from catches, counting fences, and dam passage. Relevant to this population, Bailey et al. (2025) (their appendix table 5) identified large proportions of ocean age 1 spawners (jacks) in 2019, 2007, 2021, and 2003; respectively 55.8%, 48.2%, 41.9%, and 38.3%. These years are all low runs ($< 2{,}000$), and do not have unusual entry timing, entry spread, or exit spread.

6.6. Multiple rivers

This analysis of multiple years together might be extended to include multiple rivers if spawning groups in those rivers covary consistently as parts of population or possibly a conservation unit (metapopulation). As an equation,

$$N_{tjk} = N_j P_k \ (T_0[\mu, \sigma]_t + T_j[\mu, \sigma]_t + T_k[\mu, \sigma]_t) + \epsilon_{tjk}$$

where N_{tjk} is the number of spawners entering river k on day t in year j, estimated with error ϵ from the proportion P_k of all spawners that year N_j that enter river k according to how the background temporal pattern T_0 is affected by the pattern for that year T_j and that river T_k. Terms $T_x[\ ]_t$ are temporal distributions, with μ_x and σ_x for timing and spread if Gaussian.

6.7. Residuals: lognormal versus negative binomial

> Weights are changed by log transform and/or choice of residuals model. Do you have anything to say about that changes results, or advice to users of this model? If no, delete this section.

Fitting a model to the log of abundance observations changes the relative importance (effect) of the data points, it emphasizes the contribution to the sum of squares from the residuals at small abundance estimates compared to large. The transform involves asking (a) are small abundance estimates more, or less, precise than large? and (b) which abundances are more important to estimate accurately? Minimizing residuals from log values, as done here, implies sampling error is proportional to the observation (to the true abundance): it is multiplicative as in *% accuracy,* rather than additive. If the precision of each abundance estimate can be quantified, w, then a weighted regression isappropriate $\hat{y} = \mathcal{N}(a + bx, \sigma\ w)$. Note that ordinal weights (ranked, not quantified) can be used by fitting an additional parameter, $\hat{y} = \mathcal{N}(a + bx, \sigma\ w^{-\lambda})$.

6.8. Further sources of uncertainty

> Again, if you want to say something about methods, that will be valuable. Else, delete this section.

Observer accuracy is a calibration problem, recognized as efficiency, the fraction of fish observed (bias), but includes uncertainty. This varies by method. Comparison of observers may involve factors including time of day, weather conditions, and fish density. For instance, counts while snorkeling are not possible during high flows or turbid water, and are truncated at high spawner densities.

6.9. Value of surveys for newly dead spawners

> Awaits comparisons from simulations. Could be important.
> I think readers of this paper would benefit from your experience, discovering counts of dead spawners were misleading, dead pitch events were few compared to boat surveys of live.

7. Conclusions

outline: Overall: This new approach delivered more accurate runs and estimated ecologically important variables. Runs – estimates with precision. Accuracy is better than AUC. Newly Dead – these data would have been informative. Residence * precise, CV 5%. * Did this identify an important bias in AUC? * Did not vary greatly between years (unexpected.) Timing, Spread – precise estimates, opportunity to add more ecology.

1. Residence time, a key parameter, was successfully estimated by taking advantage of newly dead spawners surveyed in multiple years spawner surveys.
   The value obtained is twice the assumed value, suggesting spawner abundance from AUC was previously overestimated.
2. The fraction of dead spawners observable (not drifted) was estimated as ~2/3, but refinement is warranted.
3. more
4. more

8. Recommendations

We propose two groups of suggestions.

As a generalization, salmon datasets are not maintained following best practices (Diack et al. (2024)). As we discovered during this analysis, datasets from spawner surveys warrant attention to:

1. Documenting survey methods, including data processing.This is essential to avoid erroneous assumptions by analysts. We point to the website Monitoring Resources maintained by the **Pacific Northwest Aquatic Monitoring Partnership.**
2. Datasets that are F.A.I.R. (Wilkinson et al. (2016)). Cataloging the existence of proprietary datasets is rewarded from judicious sharing.

Improving the accuracy of spawner run size estimates appears possible by using more of the information collected in spawner surveys but not used. This model can be expanded with further ecological realism. Simulations showed how surveys of newly dead spawners are useful to that end. With the caveat that adding complications and noisy datasets will have diminishing returns, we foresee including:

1. Habitat variables (river flow, temperature,) applied to improve the spawner entry and models; likely asymmetric or discontinuous in response to flow events;

2. Covariates from the biometrics of spawners: age, sex, age*sex, genomics, and genetics;
3. Including correlations between annual parameters (noted) might improve accuracies;
4. Temporal trends (runs, timings) are valid covariates *per se*, but ideally related to habitat variables;
5. Surveys in a year are autocorrelated, an important consideration; and
6. Sockeye spawners observed in specific year produce spawners returning three to six years later; information from a salmon's life-cycle is not extraneous to spawner surveys.

9. Acknowledgements

Diligence, care, and persistence by survey field teams from the Okanagan Nations Alliance (ONA, syilx.org) resulted in a spawners dataset based on unusually thorough and consistent methods; their commitment to *cause to come back* is inspiring. This paper commemorates accomplishments of the late Dr. Kim Hyatt who designed, implemented, and supported this survey program. We thank Karilyn Alex (ONA) and Collette Louie (Osoyoos Indian Band, oib.ca) for access and permission regarding this proprietary data, possible through the data sharing agreement of the Canadian Okanagan Basin Technical Working Group (COBTWG). We thank Howard Stiff and Athena Ogden (Pacific Biological Station, Fisheries and Oceans Canada) and Elizabeth Ng (Four Peaks Environmental Ltd.) for the work required to reconcile and summarize 25 years of field notes and spreadsheets into the quality controlled and standardized dataset we received.

* I can't find a reference to the data-sharing agreement *per se*.

Dr. James R. Irvine (DFO Pacific, retired) improved text based on his extensive knowledge of salmon surveys and related analyses.

9.1. Competing Interests

The authors declares no competing interests.

9.2. Funding Statement

The authors declares no specific funding for this work.

9.3. Data Availability

The data used in this study are the property of the Okanagan Nations Alliance. Please contact them directly concerning availability and acknowledgements: Sylix.org.

Figures

not used

Tables

not used

10. Supplementary Materials

Our selected model (m7), which uses a time-invariant $\mu^{\text{exit lag}}$, is specified as:

$$
\begin{aligned}
N_{y,d}^{\text{live}} &\sim \text{NegBinomial}(\mu_{y,d}^{\text{live}}, \phi^{\text{live}}) \\
\mu_{y,d}^{\text{live}} &= \exp\left(\log\left(1 + 10^4 \times p_{y,d}^{\text{arrived}} \times (1 - p_{y,d}^{\text{exited}})\right) - \log(10^4)\right) \times N_y^{\text{spawners}} \\
p_{y,d}^{\text{arrived}} &= \Phi\left(\frac{d - \mu_y^{\text{arrival}}}{\sigma_y^{\text{arrival}}}\right) \\
p_{y,d}^{\text{exited}} &= \Phi\left(\frac{d - (\mu_y^{\text{arrival}} + \mu^{\text{exit lag}})}{\sigma_y^{\text{exit}}}\right) \\
\log(N_y^{\text{spawners}}) &\sim \text{Normal}(10, 1) \\
\mu_y^{\text{arrival}} &\sim \text{Normal}(\mu_\mu^{\text{arrival}}, \mu_\sigma^{\text{arrival}}) \\
\mu_\mu^{\text{arrival}} &\sim \text{Normal}(40, 5) \\
\mu_\sigma^{\text{arrival}} &\sim \text{Exponential}(2) \\
\log(\sigma_y^{\text{arrival}} - 1) &\sim \text{Normal}(\sigma_\mu^{\text{arrival}}, \sigma_\sigma^{\text{arrival}}) \\
\sigma_\mu^{\text{arrival}} &\sim \text{Normal}(\log(7), 0.2) \\
\sigma_\sigma^{\text{arrival}} &\sim \text{Exponential}(10) \\
\log(\mu^{\text{exit lag}} - 1) &\sim \text{Normal}(\log(10), 0.3) \\
\log(\sigma_y^{\text{exit}} - 1) &\sim \text{Normal}(\sigma_\mu^{\text{exit}}, \sigma_\sigma^{\text{exit}}) \\
\sigma_\mu^{\text{exit}} &\sim \text{Normal}(\log(7), 0.2) \\
\sigma_\sigma^{\text{exit}} &\sim \text{Exponential}(10).
\end{aligned}
\tag{8}
$$

The following documents are available for public access on GitHub and Zenodo:
FIX LINKS

1. spawners.stan – Stan code for the models
2. Spawners_est.R – This is the R code to run the Stan model and plot results.
3. Newly Dead Spawners.pdf – Simulations report as PDF.

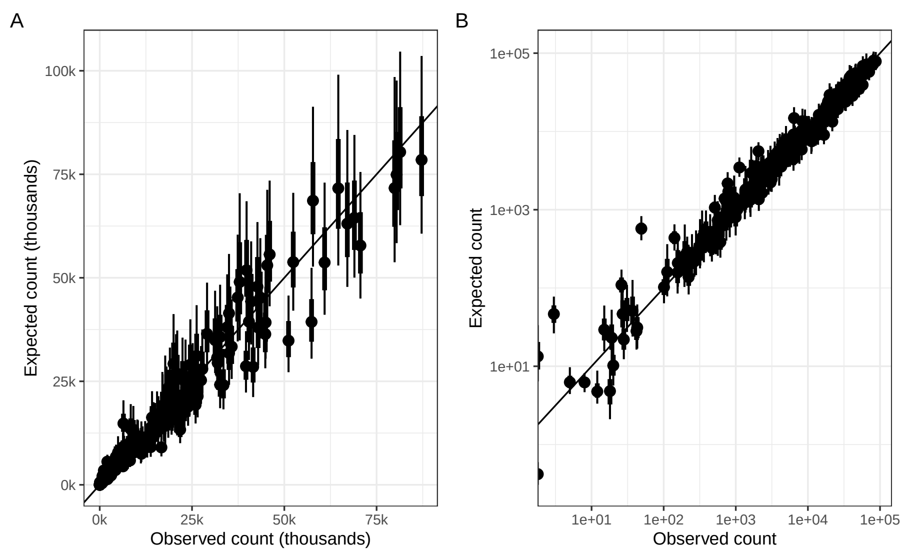

Figure S1: Observed survey counts versus predicted on linear (A) and log (B) scales. Points and error bars show the posterior median and the 66% and 95% credible intervals.

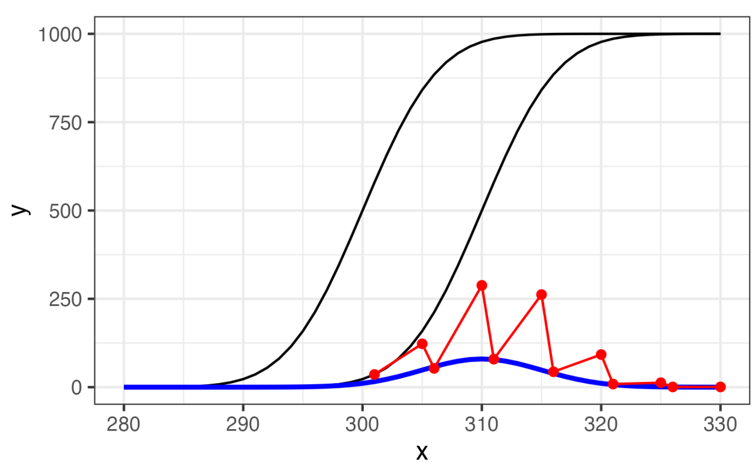

Figure S2: Newly dead spawners each day (blue) is the rate of exits. Newly dead accumulate between surveys, here with alternating 1 day and 4 day separations, leading to an irregular pattern of counts (red). Parameters as in preceding figure.

4. Newly Dead Spawners.qmd – Simulation code to describe, run, and report the model.
5. newly dead spawners.stan – Stan code for simulations.

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